

Supplementary Information for: Shared patterns of decoder drift and electrode failure suggest simple safeguards for implanted brain–computer interfaces

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Abstract

This document contains Supplementary Notes 1–3 and Supplementary Tables 1–12. Every table is generated from the released result files by `scripts/make_supp_tables.py`, so the values match the code repository exactly.

Supplementary Note 1: Label-free monitoring statistics

Before the main analyses, we asked whether label-free statistics computed on a later day can predict how well a renormalized fixed decoder will perform on that day. The pilot used the first 46–61 sessions of monkey N (January 2020 to about March 2021) and 283 session pairs at most 60 days apart. Decoders were ridge regressions on 8 lags of spike-band power ($\alpha = 0.1$), trained on the first 300 trials and tested on the rest.

We computed 14 statistics that need no labels on the later day:

- the shift in channel means and in channel standard deviations;
- the principal angles between the dominant subspaces of the two days;
- disagreement within a bootstrap ensemble of decoders;

- disagreement between decoders built on different views of the data (spike-band power versus threshold crossings, and 8 lags versus 1 lag);
- statistics of the predicted output (velocity bias, variance ratio, Wasserstein distance to the training kinematics and smoothness);
- self-consistency between predicted position and predicted velocity;
- Procrustes alignment disagreement, the Procrustes residual and the share of decoder readout outside the dominant subspace;
- neural modulation depth;
- the log ratio of impedances and the number of silent channels.

Elapsed time predicted decoder accuracy well (Spearman $\rho = -0.75$ across all pairs and -0.56 within 60 days). After controlling for log elapsed days, the partial Spearman correlation of every statistic with R^2 was at most 0.27 (output variance) and mostly below 0.1. In time-blocked cross-validated prediction of R^2 , elapsed days alone gave a mean absolute error of 0.064. The label-free statistics alone gave 0.067–0.071, and elapsed days together with the statistics gave 0.067–0.070. When the target was the gain from recalibration rather than R^2 , elapsed days gave 0.045 and the statistics gave 0.054. For detecting unexpectedly poor days (residual more than one standard deviation below the time trend), every statistic had an area under the receiver operating characteristic curve of 0.50–0.58.

The residual variation was not noise. Split-half reliability of the residuals, computed on alternating pairs of trials, was $r = 0.72$, or 0.84 after Spearman–Brown correction. Alternating pairs were needed because centre-out trials alternate between outward and return targets, which made odd/even single-trial splits negatively correlated. Adding the reference decoder’s own accuracy on the later day, a label-based measure of day difficulty, raised the explained variance from 0.29 (elapsed days alone) to 0.56. Day-to-day performance beyond the time trend is therefore real and reliable, but none of the label-free statistics we tried captured it. Abrupt faults on single channels remain detectable by construction, and such faults may still justify monitoring.

Supplementary Note 2: Simulator

The simulator (`ibci.sim`) is a feature-level generative model of chronic change. It transforms a real base session into a simulated session Δt days later, so that behaviour and latent dynamics stay realistic. Its components are listed in Supplementary Table 11 and defined in the Methods. Two further components, heavy-tailed gain jumps and random walks of raw offsets and scales, are included for decoders that are not renormalized. They do not affect renormalized decoders and were not calibrated.

Calibration. The simulated ladder must be scored with exactly the same correction code and regularization as the real ladder. An earlier calibration used regularization 100–1,000 times stronger than the values chosen on real data. That handicapped the simulated corrections, and the optimizer compensated with distorted parameters (loss 0.663 versus 0.074 after the fix). We also restricted silencing to channels that were active in the base session, because most channels are already silent on any given day.

Validation. The simulator calibrated on days before 700 was evaluated on base sessions after day 700. The comparison covered all rungs of the ladder and labelled-data budgets of 20, 50 and 100 trials. A simulator with all drift switched off served as a reference. The calibrated simulator was closer to the real data in every group (median absolute error 0.12, 0.06 and 0.11 for fitted rungs, unfitted rungs and unfitted budgets; no-drift reference 0.37, 0.27 and 0.39). Its systematic miss, faster decay than observed in the later years, led to the separate late-period calibration in Supplementary Table 11.

Intended uses and limits. The simulator is meant for stress-testing decoders and label-free stabilizers under controlled drift and electrode failure, and for comparing recalibration policies. It is calibrated on one monkey with Utah arrays, spike-band power and a finger task. Its mechanisms reproduce the measured statistics, which does not show that they are the true physical mechanisms. Users can calibrate it to other datasets with `scripts/calibrate_sim.py`.

Supplementary Note 3: Analysis choices that were corrected during the study

We report three analysis choices that we revised, because each could mislead similar studies.

A full input remap is not a structural test. A free $C \times C$ linear map in front of a frozen decoder appears to test whether drift is a “remixing” of channels. With a readout of rank at most 32 (8 lags by 4 outputs) and 96 channels, however, the map can reproduce any decoder of the same form. Its recovery therefore matches retraining and says nothing about the structure of drift. We used it only as a calibration target for the simulator.

Recovery from re-learning a few channels reflects decoder importance. Re-learning the weights of k channels recovered much of the loss when the channels were those the old decoder relied on most. Channels chosen by label-free change did worse (Supplementary Table 9). The recovery therefore reflects which channels matter to the decoder, not evidence that drift is concentrated on a few electrodes.

Small-budget recalibration must shrink the intercept. Refitting the intercept freely with 10–20 trials shifted the output offset and made recalibration harmful in many pairs. Shrinking the intercept toward the old decoder, together with the weights, removed most of this harm (Supplementary Table 8).

Supplementary Table 1 Datasets. Session counts are per analysis. Some human sessions contribute to more than one analysis.

Individual	Source	Arrays and site	Signal	Task	Sessions (span)
Monkey N	DANDI 001201 (LINK)	2 Utah, motor cortex	SBP, TC	finger flexion	312 on 303 days (3.4 years)
Monkey C	DANDI 000688	Utah, motor/premotor	sorted spikes	reaching	68 (3.0 years)
Monkey M	DANDI 000688	Utah, motor/premotor	sorted spikes	reaching	28 (1.5 years)
Human T6	Dryad x0k6djj1h	1 Utah, motor cortex	SBP	closed-loop cursor	124 (3.1 years)
14 humans	Dryad x0k6djj1h	20 Utah arrays	TC rates, impedance	research sessions	2,289 (≤ 7.3 years)
T11, T5	Dryad n2z34tn5s (MINDFUL)	motor cortex	SBP (T11), TC (T5)	closed-loop cursor	15 and 6 days

SBP, spike-band power; TC, threshold crossings.

Supplementary Table 2 Correction ladder by individual and gap. Median decoding R^2 for each rung, and median retention of the renormalized fixed decoder (renormalized R^2 divided by reference R^2) with 95% cluster-bootstrap confidence intervals. Each training session contributes at most one pair per gap. Supervised rungs used 300 labelled trials.

Gap (days)	Pairs	Reference	None	Mean update	Renorm.	Aligned	Stable aligned	Gains	Ridge to prior	Retention [95% CI]
<i>Monkey N</i>										
1	30	0.29	-2.13	0.14	0.20	0.19	0.17	0.22	0.31	0.73 [0.62, 0.78]
7	48	0.29	-1.60	0.07	0.15	0.12	0.09	0.17	0.30	0.53 [0.43, 0.58]
30	61	0.28	-2.57	-0.04	0.05	0.03	0.02	0.11	0.29	0.19 [0.12, 0.35]
120	70	0.29	-2.97	-0.04	0.01	-0.01	-0.02	0.08	0.29	0.04 [-0.06, 0.13]
480	50	0.29	-1.61	-0.03	-0.10	-0.11	-0.12	0.06	0.29	-0.36 [-0.43, -0.24]
<i>Monkey C</i>										
1	42	0.26	0.07	0.16	0.17	0.14	-0.11	0.21	0.26	0.71 [0.63, 0.77]
7	34	0.32	-0.06	0.18	0.20	0.17	-0.10	0.26	0.32	0.56 [0.44, 0.64]
30	13	0.42	-0.26	0.09	0.09	0.09	0.03	0.14	0.43	0.25 [0.20, 0.45]
120	18	0.17	-0.96	-0.03	-0.02	-0.07	-0.26	0.07	0.18	-0.09 [-0.27, 0.07]
480	38	0.14	-0.96	-0.24	-0.09	-0.24	-0.36	0.02	0.11	-0.27 [-0.61, -0.20]
<i>Monkey M</i>										
1	14	0.36	0.14	0.22	0.26	0.23	-0.04	0.29	0.36	0.77 [0.71, 0.84]
7	6	0.27	0.03	0.13	0.16	0.12	-0.15	0.19	0.27	0.61 [0.51, 0.71]
30	6	0.43	-0.35	0.19	0.25	0.25	0.21	0.29	0.44	0.54 [0.46, 0.65]
120	9	0.25	-0.28	-0.01	0.01	-0.00	-0.26	0.07	0.22	0.04 [-0.11, 0.15]
480	9	0.35	-0.30	-0.05	0.01	-0.07	-0.16	0.07	0.35	0.02 [-0.54, 0.10]
<i>Human T6</i>										
1	24	0.28	-0.70	0.07	0.16	0.15	0.15	0.26	0.28	0.73 [0.41, 0.88]
7	64	0.21	-0.52	0.01	0.11	0.08	0.09	0.18	0.21	0.66 [0.51, 0.86]
30	91	0.23	-1.08	-0.00	0.15	0.14	0.14	0.20	0.23	0.71 [0.62, 0.82]
120	109	0.23	-1.15	-0.04	0.11	0.09	0.09	0.21	0.25	0.51 [0.32, 0.61]
480	81	0.29	-2.27	-0.01	0.10	0.10	0.08	0.26	0.29	0.30 [0.17, 0.41]

Supplementary Table 3 Recurrent versus linear decoder (monkey N, 79 matched pairs). Median reference R^2 (decoder trained on the test day), retention of the renormalized fixed decoder and retention after re-learning channel gains.

Gap (days)	Pairs	Reference R^2		Retention, renormalized		Retention, gains	
		Ridge	LSTM	Ridge	LSTM	Ridge	LSTM
1	8	0.25	0.34	0.59	0.56	0.61	0.88
7	13	0.29	0.41	0.48	0.55	0.52	0.76
30	20	0.30	0.41	0.40	0.45	0.45	0.74
120	22	0.31	0.44	0.20	0.19	0.32	0.68
480	16	0.30	0.42	-0.14	-0.11	0.20	0.39

Supplementary Table 4 Offline monitoring, monkey N (537 pairs). Spearman correlation with decoder R^2 , raw and partial given log elapsed days, and time-blocked cross-validated error of predicted R^2 . “Days and divergence” uses log days together with the neural and output divergences.

Predictor	Raw ρ	Partial ρ
Elapsed days (log)	-0.67	-
Neural divergence	-0.34	-0.16
Output divergence	-0.22	-0.16
Sum	-0.36	-0.17
Cross-validated error of predicted R^2 (mean absolute error)		
Days only	0.085	
Divergence only	0.114	
Days and divergence	0.090	

Supplementary Table 5 Closed-loop monitoring in the public MINDFUL data. Pearson correlation of angular error with elapsed days and with each divergence (raw and partial given days), and leave-one-day-out mean absolute error (degrees) of predicted angular error from elapsed days alone, with neural or output divergence added, or from output divergence alone.

Participant	Days	Windows	$r(\text{error, days})$	Neural divergence		Output divergence		Prediction error (°)			
				Raw	Partial	Raw	Partial	Days	+ neural	+ output	Output only
T11	15	145	0.84	0.82	0.37	0.81	0.73	17.1	17.5	11.9	12.8
T5	6	73	0.17	0.50	0.49	0.79	0.78	32.9	38.3	20.8	15.0

Supplementary Table 6 Regularization by the 1% rule. The rule chose the largest ridge penalty whose median same-day R^2 was within 1% of the best. Columns give median R^2 with $\alpha = 0.1$ and with the chosen α , the median paired gain with 95% cluster-bootstrap confidence interval, and the share of training sessions whose mean gain was positive.

Individual	Chosen α	Gap (days)	Pairs	R^2 , $\alpha = 0.1$	R^2 , rule	Gain	[95% CI]	Share positive
Monkey N	10^4	same day	40	0.30	0.30	0.003	[0.001, 0.006]	0.72
Monkey N	10^4	1	11	0.27	0.28	0.021	[0.008, 0.035]	1.00
Monkey N	10^4	7	24	0.14	0.18	0.035	[0.026, 0.041]	0.92
Monkey N	10^4	30	28	0.10	0.14	0.042	[0.038, 0.052]	1.00
Monkey N	10^4	120	35	0.04	0.10	0.056	[0.050, 0.083]	0.97
Monkey N	10^4	480	28	-0.10	-0.00	0.090	[0.082, 0.112]	1.00
Monkey C	10^4	same day	40	0.21	0.22	0.002	[0.001, 0.003]	0.72
Monkey C	10^4	1	26	0.16	0.18	0.008	[0.005, 0.017]	0.88
Monkey C	10^4	7	22	0.20	0.21	0.012	[0.006, 0.019]	0.77
Monkey C	10^4	30	9	0.10	0.10	0.001	[-0.002, 0.011]	0.67
Monkey C	10^4	120	12	-0.01	0.01	0.031	[0.016, 0.045]	1.00
Monkey C	10^4	480	25	-0.10	-0.07	0.032	[0.014, 0.036]	1.00
Monkey M	10^3	same day	22	0.35	0.35	0.000	[0.000, 0.001]	0.73
Monkey M	10^3	1	14	0.26	0.26	0.001	[0.000, 0.001]	0.79
Monkey M	10^3	7	6	0.16	0.17	0.000	[-0.000, 0.002]	0.67
Monkey M	10^3	30	6	0.25	0.25	0.001	[0.000, 0.012]	1.00
Monkey M	10^3	120	9	0.01	0.01	0.001	[0.000, 0.002]	1.00
Monkey M	10^3	480	9	0.01	0.01	0.001	[0.000, 0.001]	1.00
Human T6	10^4	same day	40	0.22	0.24	0.008	[0.003, 0.013]	0.95
Human T6	10^4	1	6	0.13	0.17	0.011	[0.002, 0.075]	1.00
Human T6	10^4	7	23	0.10	0.14	0.025	[0.016, 0.032]	1.00
Human T6	10^4	30	32	0.16	0.17	0.016	[0.013, 0.030]	0.97
Human T6	10^4	120	38	0.13	0.17	0.018	[0.009, 0.030]	0.95
Human T6	10^4	480	27	0.11	0.12	0.017	[0.005, 0.027]	0.89

Supplementary Table 7 Recalibration with few labelled trials (monkey N, 86 pairs). Median retention of the renormalized fixed decoder (no labels), and for each number of labelled trials the median retention of ridge to prior / retraining from scratch / channel gains. Regularization chosen by cross-validation on the labelled trials.

Gap (days)	Pairs	No labels	$n = 10$	$n = 20$	$n = 50$	$n = 100$	$n = 300$
1	7	0.64	0.47 / -0.18 / 0.02	0.63 / 0.22 / 0.50	0.75 / 0.55 / 0.64	0.83 / 0.70 / 0.69	1.03 / 1.02 / 0.73
7	18	0.48	0.51 / 0.17 / 0.38	0.58 / 0.39 / 0.43	0.77 / 0.68 / 0.54	0.89 / 0.81 / 0.61	1.02 / 1.01 / 0.54
30	20	0.15	0.21 / -0.07 / 0.05	0.27 / 0.25 / 0.09	0.60 / 0.61 / 0.25	0.79 / 0.80 / 0.29	1.02 / 1.02 / 0.27
120	26	0.13	0.31 / 0.24 / -0.02	0.44 / 0.43 / 0.06	0.67 / 0.66 / 0.24	0.82 / 0.83 / 0.19	1.01 / 1.01 / 0.30
480	15	-0.21	0.19 / 0.19 / -0.27	0.35 / 0.42 / -0.07	0.60 / 0.64 / 0.16	0.82 / 0.83 / 0.09	1.01 / 1.01 / 0.27

Supplementary Table 8 Choosing the shrinkage strength. Share of harmful recalibrations (more than $0.01 R^2$ below the renormalized fixed decoder) when the strength of ridge to prior is chosen by cross-validation on the available trials or from past sessions, with two-sided Fisher’s exact test, and median retention for both policies and for the best strength in hindsight (an upper bound, not a usable policy).

Trials	Pairs	Harmful (%)			Median retention		
		Cross-validation	History	P	Cross-validation	History	Hindsight
10	86	10.5	1.2	0.018	0.31	0.37	0.40
20	86	8.1	0.0	0.014	0.43	0.44	0.48
50	86	1.2	0.0	1.0	0.64	0.63	0.68
100	86	0.0	1.2	1.0	0.82	0.81	0.83
300	86	0.0	0.0	1.0	1.02	1.01	1.02

Supplementary Table 9 Re-learning 32 channels only. Median retention of a full refit (ridge to prior, strength from past sessions) and median paired difference in R^2 from it, with 95% cluster-bootstrap confidence intervals, when only 32 channels are re-learned. Channels were chosen by decoder importance, by label-free change, by importance-weighted change, or at random.

Trials	Pairs	Full refit	Importance	Label-free change	Weighted change	Random
10	86	0.37	0.014 [0.004, 0.028]	-0.012 [-0.017, -0.009]	0.011 [-0.003, 0.018]	-0.019 [-0.023, -0.015]
20	86	0.44	0.022 [0.011, 0.028]	-0.015 [-0.027, -0.011]	0.014 [0.001, 0.025]	-0.032 [-0.038, -0.029]
50	86	0.63	0.020 [0.008, 0.029]	-0.031 [-0.043, -0.026]	0.007 [0.000, 0.017]	-0.064 [-0.070, -0.053]
100	86	0.81	-0.004 [-0.008, -0.000]	-0.042 [-0.054, -0.034]	-0.010 [-0.020, -0.003]	-0.085 [-0.109, -0.065]

Supplementary Table 10 Electrode failure statistics per array. Part., participant; Sess., sessions. Yield is the percentage of electrodes with a threshold-crossing rate of at least 2 Hz, as the median of the first / last three sessions. h_{off} and h_{on} are the active-to-silent and silent-to-active switching rates per 1,000 days. Rev./sil. counts revived and silenced initially active electrodes. Kurt., excess kurtosis. Kurtosis is computed on session-to-session changes in log rate after removing session-wide changes. Imp. ρ is the Spearman correlation of the array median with day. Edge – int. gives the median edge minus median interior slope of log rate per year, and P is the one-sided Mann–Whitney test that edge electrodes decline faster. For monkey N, values are given for the lateral and medial arrays where they differ.

Part.	Array	Sess.	Span (d)	Yield (%)	h_{off}	h_{on}	Rev./sil.	Kurt.	Imp. ρ	Edge – int.	P
A1	single	48	252	81 / 55	16.3	27.1	44/68	2.9	–0.27	0.69	0.322
S1	single	40	337	3 / 1	46.6	10.7	1/4	0.7	–	–0.14	0.017
S2	single	32	458	1 / 6	24.2	10.9	0/1	–	–	0.38	0.925
S3	single	207	1,954	84 / 10	14.9	3.3	68/88	3.9	–0.91	–0.10	0.040
T1	single	41	277	65 / 86	2.6	27.1	11/17	7.8	–0.93	–0.18	0.379
T10	MFG	104	329	96 / 86	8.6	54.1	56/59	9.8	–0.70	0.27	0.877
T10	dPCG	104	329	21 / 44	7.7	9.1	26/31	3.2	–0.21	–0.84	0.006
T11	lateral	292	1,677	20 / 10	7.0	1.5	16/24	5.5	–0.95	–0.08	0.025
T11	medial	292	1,677	38 / 64	3.7	6.3	35/35	4.1	–0.99	–0.12	0.039
T2	single	176	914	57 / 14	22.4	3.4	50/67	4.1	–0.98	0.50	0.966
T3	single	87	399	0 / 2	34.3	4.0	0/0	–	–0.98	–0.02	0.131
T5	lateral	438	2,665	60 / 30	2.4	1.2	41/59	4.4	–0.94	–0.05	0.055
T5	medial	438	2,665	51 / 40	2.9	2.7	54/59	2.4	–0.90	–0.03	0.271
T6	single	190	1,184	16 / 2	11.0	3.5	12/22	2.7	–0.97	–0.11	0.059
T7	lateral	84	521	24 / 31	20.1	21.7	23/28	1.9	–0.83	–0.05	0.161
T7	medial	84	521	36 / 49	8.7	11.8	34/42	1.5	–0.91	–0.81	<0.001
T8	lateral	329	1,061	11 / 38	1.1	1.2	12/12	6.7	–0.92	0.29	0.986
T8	medial	329	1,061	55 / 41	5.2	2.9	64/70	3.0	–0.92	0.27	0.805
T9	lateral	221	731	14 / 16	18.7	7.5	13/17	4.3	0.26	–0.98	<0.001
T9	medial	221	731	21 / 44	13.1	6.6	24/27	2.1	–0.96	–0.22	0.028
Monkey N	both	303	1,242	38 / 19	3.1	0.8	23/31	10.2	–0.88	–0.15 / –0.09	<0.001 / 0.046

Supplementary Table 11 Simulator parameters.

Values calibrated on monkey N sessions before day 700 (early) and from day 700 onwards (late). Switching rates were measured from electrode activity in each period rather than calibrated. Fixed parameters: local mixing share $\beta = 0.5$, turnover concentration 5, 20 latent dimensions and silent-signal fraction $\kappa = 0.37$.

Parameter	Unit	Early	Late
Session-to-session mixing s.d.	1	0.959	0.768
Slow mixing s.d. (asymptote)	1	1.478	1.317
Slow mixing time constant	days	15	12
Turnover fraction, next session	1	0.042	0.032
Turnover fraction, asymptote	1	0.120	0.127
Turnover time constant	days	360	527
Active to silent rate	per day	0.0038	0.0034
Silent to active rate	per day	0.0018	0.0017
Calibration loss		0.074	0.063

Supplementary Table 12 Training on simulated futures. Median R^2 on real later sessions of monkey N for ridge decoders trained on sessions after day 700 (simulator calibrated before day 700). Base, $\alpha = 0.1$; stronger ridge, $\alpha = 10^4$ chosen on the calibration period; perturbation, random channel dropout and gain noise; simulator, eight simulated futures; simulator and ridge, both; several days, real data from the three previous sessions; reference, decoder trained on the test day.

Gap (days)	Pairs	Base	Stronger ridge	Perturbation	Simulator	Simulator and ridge	Several days	Reference
1	9	0.21	0.22	0.20	0.20	0.20	0.23	0.29
7	20	0.18	0.21	0.18	0.19	0.20	0.23	0.34
30	18	0.08	0.13	0.08	0.09	0.10	0.15	0.28
120	18	0.07	0.12	0.07	0.07	0.08	0.14	0.30
240	12	0.05	0.10	0.06	0.08	0.08	0.11	0.29